

A probabilistic characterization of the health benefits of reducing methyl mercury intake in the United States.

We developed a probabilistic model to characterize the plausible distribution of health and economic benefits that would accrue to the U.S. population following reduction of methyl mercury (MeHg) exposure. MeHg, a known human developmental neurotoxicant, may increase fatal heart attack risks. Model parameters reflect current understanding of the relationships between MeHg intake, health risks, and societal valuation of these risks. The expected monetary value of the annual health benefits generated by a 10% reduction in U.S. population exposure to MeHg for one year is \$860 million; 80% of this is associated with reductions in fatal heart attacks and the remainder with IQ gains. The plausible distribution of the benefits is quite broad with 5th and 95th percentile estimates of approximately \$50 million and \$3.5 billion, respectively. The largest source of uncertainty is whether epidemiological associations between MeHg exposure and fatal heart attacks reflect causality. The next largest sources of uncertainty concern the slope of the relationship between maternal MeHg exposure and reduced intelligence among children and whether this relationship exhibits a threshold. Our analysis suggests that the possible causal relationship between MeHg exposure and fatal heart attacks should be better characterized, using additional epidemiological studies and formally elicited expert judgment.